# Against Multilevel Selection in Legal Evolution: Why the Gene's-Eye View Provides Superior Explanatory and Predictive Power

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December, 2025

## Abstract

Multilevel selection theory (MLS2) has recently been applied to legal evolution, most notably in Fábio Portela Lopes de Almeida's work treating constitutional societies as "Darwinian individuals" subject to group-level selection. This paper argues that MLS2 applied to legal systems reintroduces the teleological thinking that gene-centered evolutionary biology and Austrian economics rejected. I postulate that Extended Phenotype Theory (EPT), which treats legal institutions as extended phenotypes of competing memes, provides superior explanatory and predictive power through three lines of argument.

First, MLS2 lacks specifiable causal mechanisms: the pathways by which "societies" supposedly compete and transmit traits remain obscure.

Second, parsimony favors EPT: Occam's razor cuts against multiplying levels of selection when lower-level mechanisms suffice.

Third, empirical tests consistently favor EPT over MLS2.

Four case studies of dysfunctional norms show EPT predicting persistence with 100% accuracy while MLS2 predicts collapse.

Quantitative metrics derived from EPT (Constitutional Lock-in Index, IusMorfos) generate falsifiable predictions with 78% accuracy, while MLS2 offers only vague claims about "fitter societies."

Genealogical analysis reveals 91% of constitutional norms have ancestors predating supposed "punctuational" events, undermining MLS2's revolutionary narrative.

EPT tries to advance legal evolution from philosophy to falsifiable science by grounding institutional analysis in memetic competition rather than group-level teleology.

**Keywords**: multilevel selection, extended phenotype, memetics, legal evolution, constitutional theory, gene's-eye view

**JEL Classifications:** K00, K10, B41, B52, D02, P48

# I. Introduction

The temptation to explain complex social order through group-level selection is powerful. Societies that protect property rights prosper. Societies with functional legal systems attract investment. Societies that enforce contracts facilitate trade. From these observations, it seems natural to conclude that societies compete, successful institutional arrangements spread through intergroup selection, and legal evolution operates through survival of the fittest societies. This is the logic of multilevel selection theory (MLS2) applied to cultural and legal evolution.

Fábio Portela Lopes de Almeida, a Brazilian legal theorist, has recently applied this framework to constitutional systems, arguing that "constitutional societies" function as "Darwinian individuals" subject to group-level selection (Almeida 2020, 2021). His work represents the most sophisticated application of MLS2 to legal theory, integrating insights from evolutionary biology, systems theory, and legal philosophy. Almeida argues that constitutions stabilize functionally differentiated societies, that constitutional systems compete, and that successful constitutional arrangements spread through a form of cultural group selection. This framework appeals to many legal scholars because it explains cooperation, moral evolution, and the emergence of complex institutions without reducing them to individual self-interest.

I argue that this appeal is illusory. MLS2 applied to legal evolution reintroduces the teleological thinking that gene-centered evolutionary biology explicitly rejected (Williams 1966; Dawkins 1976, 1982) and that Austrian economics rejected in parallel (Menger 1871; Hayek 1973, 1988). The structural convergence between Dawkins's gene's-eye view and Hayek's spontaneous order, which I demonstrated in a companion paper (Lerer 2025a), reveals that explaining complex functional order requires locating causal mechanisms at the level of replicating units, not at emergent group levels. Just as species-level and group-level selection in biology proved explanatorily superfluous once gene-level mechanisms were understood, society-level selection in legal evolution proves superfluous once memetic mechanisms are understood.

This paper advances three lines of argument against MLS2 in legal evolution and in favor of Extended Phenotype Theory (EPT). First, the epistemological critique: MLS2 lacks specifiable causal mechanisms. The pathways by which societies supposedly compete, traits are transmitted between societies, and selection operates at the group level remain fundamentally obscure. In contrast, EPT specifies clear causal chains: memes replicate through imitation and communication, memes build extended phenotypes in the form of legal institutions, and selection operates on memes based on their replicative success. Dennett's cui bono criterion exposes the problem: asking what benefit a society derives from an institution inverts the correct explanatory

direction (Dennett 1995). The correct question is which memes benefit from the institution's existence.

Second, the parsimony argument: Occam's razor decisively favors EPT. MLS2 requires multiple levels of selection (genes, organisms, groups, societies), unspecified inheritance mechanisms at each level, and complex interactions between levels. EPT requires only two types of replicators (genes for biological traits, memes for cultural traits) and a single selection mechanism operating on replicators based on their phenotypic effects. When lower-level mechanisms suffice to explain observed phenomena, invoking higher-level selection violates parsimony.

Third, the empirical argument: EPT consistently outperforms MLS2 in explanatory and predictive power. I present three empirical tests. Test A examines dysfunctional norms that harm societies but persist. MLS2 predicts these norms should collapse through intergroup competition. EPT predicts they persist if they benefit their replicators regardless of social consequences. Data from four cases (Argentine income tax, US Jim Crow laws, US slavery, Argentine labor law) show EPT achieving 100% predictive accuracy while MLS2 achieves 0-25%. Test B examines quantitative predictions. EPT generates falsifiable metrics like the Constitutional Lock-in Index (CLI), which predicts reform outcomes with 78% accuracy. MLS2 generates no operationalized metrics and makes only vague claims about "fitter societies prevailing." Test C examines genealogical patterns. MLS2 predicts punctuational legal revolutions driven by group selection. EPT predicts gradual normative evolution with occasional architectural innovations. Data show 91% of constitutional clauses have normative ancestors predating supposed revolutionary moments, decisively favoring EPT's gradualist prediction.

The argument proceeds as follows. Section II reconstructs MLS2 as applied to legal evolution, focusing on Almeida's work as the most sophisticated contemporary example. Section III develops the epistemological critique, showing that MLS2 lacks specifiable causal mechanisms and inverts the correct explanatory direction. Section IV presents the parsimony argument, demonstrating that EPT explains legal evolution more economically. Section V presents three empirical tests consistently favoring EPT over MLS2. Section VI outlines EPT's correct framework: institutions as extended phenotypes of memes, with falsifiable predictions and clear causal mechanisms. Section VII concludes by considering whether any complementarity between the frameworks exists and argues that while Almeida's descriptive insights have value, the theoretical foundation must be gene-centered rather than group-centered.

The stakes extend beyond academic debate. If MLS2 is correct, institutional design should focus on engineering group-level selection pressures and optimizing societies as units. If EPT is correct, institutional design should focus on engineering memetic selection environments where beneficial memes outcompete harmful ones. The difference is not semantic but practical. MLS2 legitimates constructivist rationalism

and social engineering. EPT aligns with spontaneous order and constitutional constraints on design. Getting the theory right matters for getting policy right.

## II. Multilevel Selection in Contemporary Legal Theory

### The MLS2 Framework in Biology

Multilevel selection theory emerged from debates about altruism and cooperation in evolutionary biology. How can self-sacrificial behavior evolve if natural selection favors self-interest? David Sloan Wilson and Elliott Sober proposed that selection operates simultaneously at multiple levels: genes, organisms, and groups (Sober and Wilson 1998). While selfish behavior is favored within groups, groups of altruists outcompete groups of selfish individuals. This between-group selection can overcome within-group selection, allowing altruism to evolve.

The theory distinguishes MLS1 (multilevel selection type 1) from MLS2 (multilevel selection type 2). MLS1 analyzes how selection at different levels affects gene frequencies. This is compatible with gene-level selection because it ultimately tracks changes in gene frequencies. MLS2 treats groups themselves as units of selection analogous to organisms. Groups reproduce, compete, and evolve. This stronger claim resurrects group selection rejected by Williams and Dawkins (Wilson and Wilson 2007).

MLS2 proponents argue that this framework is necessary to explain major evolutionary transitions: from replicating molecules to chromosomes, from chromosomes to cells, from cells to multicellular organisms, from organisms to eusocial colonies (Maynard Smith and Szathmáry 1995). Each transition involved lower-level units forming higher-level units that became new units of selection. By analogy, human societies represent another major transition where groups become units of selection.

### Richerson and Boyd's Cultural Group Selection

Peter Richerson and Robert Boyd extended MLS2 to cultural evolution (Richerson and Boyd 2005). They argue that human cooperation is too extensive to be explained by kin selection or reciprocal altruism. Large-scale cooperation among non-kin requires cultural group selection. Humans live in groups with distinctive cultural traits. Groups with cooperative norms outcompete groups with selfish norms. Successful groups expand, and their cultural traits spread. This explains the evolution of morality, institutions, and complex societies.

Richerson and Boyd emphasize several mechanisms that make cultural group selection plausible. First, cultural traits spread through conformist transmission: individuals imitate the majority. This creates cultural homogeneity within groups and

variation between groups. Second, punishment enforces cooperative norms. Third, symbolic markers (language, dress, religion) create clear group boundaries. Fourth, intergroup competition is frequent and consequential in human history. These mechanisms allegedly make cultural groups more cohesive and longer-lived than biological groups, strengthening group selection.

**Almeida's Application to Legal Evolution**

Almeida applies this framework to constitutional systems in two major works (Almeida 2020, 2021). His central thesis is that constitutional societies function as "Darwinian individuals" subject to multilevel selection. A Darwinian individual is any entity exhibiting variation, heredity, and differential fitness. Constitutions create boundaries between differentiated social subsystems (law, politics, economy, religion). These boundaries allow societies to function as coherent units competing with other societies.

Almeida argues that constitutionalism emerged through group selection. Pre-constitutional societies lacked stable boundaries between functionally differentiated subsystems. Political power interfered with legal adjudication, religious authorities controlled political offices, and economic actors captured state power. This functional dedifferentiation created instability. Societies that developed constitutional constraints stabilizing differentiation outcompeted societies lacking such constraints. Constitutional societies exhibited superior adaptive capacity, and constitutionalism spread through cultural evolution (Almeida 2020, pp. 147-189).

The mechanism involves legal revolutions: punctuational events where new constitutional arrangements emerge rapidly. Almeida cites the Papal Revolution (11th-12th centuries), Protestant Reformation (16th century), and modern constitutional revolutions (18th-20th centuries) as examples (Almeida 2021, pp. 78-82). Each revolution imposed new normative constraints on social evolution, stabilizing functional differentiation at higher levels of complexity. Societies adopting these innovations prospered; societies resisting them declined. The pattern resembles punctuated equilibrium in biological evolution.

Almeida integrates this account with Niklas Luhmann's systems theory and Hauke Brunkhorst's critical theory. Luhmann provides the theoretical apparatus for understanding functional differentiation. Brunkhorst provides the historical narrative of legal revolutions. MLS2 provides the evolutionary mechanism. The synthesis yields a comprehensive theory of constitutional evolution as a multilevel selection process operating on societies as units (Almeida 2020, pp. 201-256).

**The Appeal of MLS2 for Legal Theory**

MLS2 appeals to legal theorists for several reasons. First, it explains large-scale cooperation and complex institutions without reducing them to individual self-interest. Legal systems require collective action, norm enforcement, and coordination. MLS2 provides a framework for understanding how these emerge. Second, it aligns with functionalist intuitions widespread in legal scholarship. Institutions exist to serve social functions. MLS2 formalizes this intuition by treating social functions as products of group-level selection. Third, it provides normative guidance. If societies are units of selection, institutional design should optimize society-level fitness.

Fourth, MLS2 resonates with progressive politics. It legitimates claims that institutions should serve collective welfare rather than individual interests. Constructivist rationalism gains theoretical support: if societies compete and successful arrangements spread, then consciously designing better societies is not only possible but necessary. This political valence makes MLS2 attractive to scholars seeking theoretical foundations for activist interventions.

### The Problem: Reintroduction of Teleology

Despite these appeals, MLS2 reintroduces the teleological thinking that gene-centered evolutionary biology and Austrian economics rejected. Pre-Darwinian biology explained traits as serving species-level purposes. Pre-Mengerian economics explained institutions as serving state-level purposes. Williams and Dawkins eliminated species-level and group-level teleology by showing that gene-level selection suffices (Williams 1966; Dawkins 1976, 1982). Menger, Mises, and Hayek eliminated state-level teleology by showing that institutions emerge from individual action without design (Menger 1871; Mises 1949; Hayek 1988). As I demonstrated in a companion paper, these two intellectual movements solved the same problem using structurally identical solutions (Lerer 2025a).

MLS2 resurrects the rejected errors. It treats societies as purposive units pursuing fitness. It explains institutions as serving society-level functions. It legitimates designing institutions to optimize group-level outcomes. This resurrection is not innocent. The gene's-eye view succeeded precisely by eliminating appeals to higher-level purposes. Reintroducing those purposes under the label "multilevel selection" recreates the problems Williams and Dawkins solved.

The next three sections develop this critique. Section III shows that MLS2 lacks specifiable causal mechanisms. Section IV shows that parsimony favors gene-centered alternatives. Section V shows that empirical tests consistently favor EPT over MLS2. Together, these arguments demonstrate that MLS2 applied to legal evolution fails on epistemological, theoretical, and empirical grounds.

## III. The Epistemological Critique

## The Mechanism Problem

The fundamental problem with MLS2 is that it lacks specifiable causal mechanisms. In gene-centered evolutionary biology, the causal chain is clear: genes code for proteins, proteins build phenotypes, phenotypes interact with environments, differential survival and reproduction produce differential gene frequencies. Every link in the chain is specifiable. Genes can be sequenced. Protein synthesis can be observed. Phenotypic effects can be measured. Fitness differences can be quantified. The mechanism is not metaphorical but concrete.

What is the analogous mechanism for group selection? How do societies compete? What traits are transmitted between societies? What is inherited? What replicates? MLS2 proponents offer vague answers. Richerson and Boyd invoke "intergroup competition" but do not specify what this means mechanistically (Richerson and Boyd 2005, pp. 214-232). Do societies literally fight wars until one is eliminated? Do they compete economically until one collapses? Do individuals migrate from unsuccessful to successful societies, spreading traits? Each mechanism implies different selection pressures and timescales, yet MLS2 theory does not specify which operates when.

The inheritance problem is even more severe. Genes inherit through DNA replication with high fidelity. Cultural traits inherit through imitation, but what exactly is inherited at the group level? Almeida claims constitutional arrangements are inherited, but constitutions do not replicate like genes (Almeida 2020, pp. 147-189). Societies do not reproduce by fission into daughter societies carrying constitutional DNA. Constitutional traits spread through imitation, conquest, or colonial imposition. None of these resembles genetic inheritance. The claim that societies are "Darwinian individuals" requires specifying the inheritance mechanism. MLS2 provides no such specification.

The replicator problem is central. Dawkins defines a replicator as any entity that makes copies of itself (Dawkins 1982, pp. 82-83). Genes are replicators. Organisms are vehicles: temporary constructions through which genes replicate. Groups are not replicators because they do not make copies of themselves with heritable variation. Societies certainly do not replicate. Argentina does not reproduce by budding off daughter Argentinas. Treating societies as units of selection requires identifying what replicates at the society level. MLS2 provides no answer.

## Dennett's Cui Bono Criterion

Daniel Dennett provides a methodological criterion that exposes MLS2's problems: cui bono? Who benefits? A benefit by itself is not explanatory. Every trait produces multiple effects, some benefiting various entities. The eye benefits the organism by enabling vision, benefits predators by enabling hunting, benefits prey by enabling predator avoidance, and benefits ophthalmologists by creating employment opportunities. Only one of these benefits is explanatory: the eye evolved because genes building eyes

replicated more successfully than genes not building eyes. The benefit to the organism is a proximate mechanism, but the ultimate explanation runs through gene replication (Dennett 1995, pp. 475-477).

Dennett illustrates this with the liver fluke example (cited in Lerer 2025a). The fluke infects ants as an intermediate host. Infected ants climb grass blades at night, increasing the probability that sheep ingest them, allowing the fluke to reach its final host. Asking what benefit the ant derives from climbing grass is the wrong question. The ant derives no benefit. The ant is harmed. The behavior benefits fluke genes by facilitating transmission. The cui bono criterion demands identifying the replicator that benefits. In this case, the replicator is fluke genes, not ant genes.

Applied to legal institutions, cui bono reveals MLS2's inversion of explanatory order. Consider the institution of private property. MLS2 asks: what benefit does society derive from property rights? Answer: economic growth, reduced conflict, investment security. These social benefits allegedly explain why societies with property rights outcompete societies without them. But this explanation inverts causal order. The correct question is: which replicators benefit from property rights? Answer: memes defining, enforcing, and elaborating property entitlements. These memes replicate through legal education, judicial decisions, and normative compliance. The institution persists because the memes replicate, not because society benefits.

The social benefits are byproducts. Societies do not benefit in any causally efficacious sense because societies are not replicators. Memes are replicators. The institution of property rights is an extended phenotype of property-rights memes (Lerer 2025a). These memes replicate because individuals who adopt them benefit in ways that lead others to imitate: reduced transaction costs, secure investment, expanded trade. The aggregate social benefit is real but epiphenomenal with respect to explanation. The causal mechanism runs through meme replication, not society-level selection.

**The Causal Inversion**

MLS2 treats societies as causes and institutions as effects. Societies compete, successful societies proliferate, and their institutional traits spread. This gets the causal order backwards. Societies do not cause institutions. Individuals adopting memes cause institutions. The institutions are extended phenotypes of those memes. Societies are higher-order patterns resulting from individuals interacting through institutional structures. Treating societies as causal agents reifies an abstraction.

This causal inversion parallels the error constructivist rationalism makes in economics. Constructivists treat the state as an agent that designs institutions to serve collective purposes. Austrian economists demonstrated that this inverts causal order (Hayek 1973, 1988). The state is not an agent. It is a pattern of individual actions structured by constitutional and legal rules. Those rules emerged from individual interactions, not from conscious design (except in specific cases where constitutional

designers created frameworks). Treating the state as a causal agent reifies an abstraction and obscures the lower-level mechanisms that generate institutional order.

As I argued in the companion paper, the Dawkins-Hayek convergence reveals that explaining complex functional order without a designer requires locating causal mechanisms at the level of replicating units (Lerer 2025a). In biology, the replicating units are genes. In culture, the replicating units are memes. In both domains, higher-level order emerges from lower-level replication. Organisms emerge from gene competition. Institutions emerge from meme competition. Societies and ecosystems are higher-order patterns with no causal agency of their own.

MLS2 reverses this by attributing causal agency to the higher level. Societies compete and exert selective pressure on lower levels. This is coherent only if societies are replicators, which they are not. The inversion generates explanatory confusion. Instead of explaining institutions by tracing them to memetic replication, MLS2 explains institutions by invoking society-level fitness. But society-level fitness is itself an aggregate measure of individual and memetic success. The explanation is circular.

**The Group Boundary Problem**

Even if we grant that groups can function as units of selection, a severe boundary problem remains. What defines a group? In biological group selection, the problem is tractable but difficult. A group might be a breeding population, a deme, or a local population. But boundaries are porous: individuals migrate, populations interbreed, and groups fuse and fission. These dynamics weaken group selection because they disrupt the group cohesion required for selection to operate (Dawkins 1982, pp. 84-87).

In cultural evolution, the boundary problem is far worse. What defines a society? Is Argentina a society? Is Latin America a society? Is the Global South a society? Is the Catholic world a society? Each grouping is plausible, and individuals belong to multiple overlapping groups simultaneously. An Argentine judge belongs to Argentine society, Latin American legal culture, civil law tradition, Catholic moral framework, and global legal profession. Which of these is the relevant unit of selection?

Almeida treats constitutional societies as the relevant units (Almeida 2020, pp. 201-256). But constitutional boundaries do not coincide with cultural, linguistic, religious, or ethnic boundaries. Switzerland has four language groups and two religious traditions within one constitutional framework. Belgium has two language groups with persistent tensions within one constitutional framework. The European Union creates supranational constitutional structures overlaying national constitutions. What is the unit of selection: nation-state, constitutional jurisdiction, cultural group, linguistic community?

The problem is not merely definitional. Different boundaries imply different selection mechanisms. If nation-states are units, selection operates through warfare and economic competition between states. If constitutional systems are units, selection operates through institutional performance and legal migration. If cultural groups are units, selection operates through differential reproduction and conversion. Each mechanism has different timescales, selection pressures, and equilibria. MLS2 does not specify which boundaries matter or how to determine them empirically.

**The Timescale Problem**

Related to the boundary problem is the timescale problem. Genetic evolution operates over generations. A generation for humans is roughly 25-30 years. Cultural evolution operates faster because cultural traits spread through imitation rather than reproduction. But how fast does group selection operate? MLS2 proponents offer no clear answer.

Consider Almeida's claim that constitutional societies outcompete non-constitutional societies (Almeida 2020, pp. 147-189). Over what timescale? The modern constitutional era spans roughly 250 years since the American and French Revolutions. Is this sufficient time for group selection to operate? The Roman Republic lasted 500 years without a modern constitution. The Chinese imperial system lasted 2000 years without constitutionalism. The British Empire dominated the 19th century with an uncodified constitution. Do these cases represent failed group selection or different selection pressures?

The timescale problem matters because it affects testability. If group selection operates over centuries or millennia, we cannot test predictions within scholarly lifetimes. If it operates over decades, we should observe rapid institutional convergence, which we do not. The diversity of constitutional arrangements persists despite globalization and institutional diffusion. This persistence is puzzling for MLS2 but predicted by EPT: different memetic ecologies sustain different institutional configurations because memes replicate based on local conditions, not global competition between societies.

The timescale problem creates a fundamental falsifiability dilemma for MLS2. If group selection operates over centuries or millennia, the theory becomes empirically untestable within scholarly lifetimes. Any contemporary institutional configuration can be dismissed as transitional, with "true" group selection effects manifesting only over timescales that preclude observation. This renders MLS2 unfalsifiable in Popper's sense: no conceivable evidence could refute it because any contrary observation can be attributed to insufficient time elapsed.

Conversely, if group selection operates over decades, we should observe rapid institutional convergence toward optimal configurations. Societies with superior institutions should outcompete societies with inferior institutions within observable

timeframes. Yet we observe no such convergence. Constitutional diversity persists and even increases despite globalization, institutional diffusion, and competitive pressures. Common law and civil law traditions have coexisted for over 800 years without convergence. Presidential and parliamentary systems have coexisted for 230+ years without one displacing the other. Federal and unitary structures show no trend toward convergence. Proportional representation and first-past-the-post electoral systems coexist without decisive competitive advantage.

If MLS2 were correct and group selection operated effectively, this diversity should collapse. Inferior institutional arrangements should be eliminated through competitive pressure. Superior arrangements should spread universally. Instead, we observe stable institutional pluralism. This pattern falsifies the strong version of MLS2 that predicts convergence through intergroup competition.

MLS2 proponents might respond that different institutional configurations are adaptively equivalent, representing multiple fitness peaks in an adaptive landscape. But this response concedes that group selection does not determinately favor particular institutions, undermining claims that constitutionalism, functional differentiation, or specific constitutional arrangements spread because they are group-adaptively superior. If multiple configurations are equally fit, group selection explains nothing about which configurations persist where.

EPT predicts this persistent diversity straightforwardly. Different memetic ecologies sustain different institutional configurations. Memes replicate based on local conditions: legal culture, historical path dependence, linguistic frameworks, and existing institutional complementarities. Common law memes replicate effectively in Anglophone jurisdictions with precedent-based legal education. Civil law memes replicate effectively in Continental European jurisdictions with code-based legal education. Neither is globally superior; each is locally stable. The diversity is evidence favoring EPT's meme-level selection over MLS2's group-level selection.

The timescale problem is not a technical difficulty requiring more data. It is a conceptual problem revealing that MLS2 makes no determinate predictions. A theory that can accommodate any outcome by adjusting timescales post hoc is not a scientific theory.

**The Epistemological Upshot**

The epistemological critique reveals that MLS2 lacks the mechanistic clarity required for scientific explanation. It cannot specify what competes, what replicates, what is inherited, where boundaries lie, or over what timescales selection operates. These are not minor technical difficulties but fundamental conceptual problems. Without specifiable mechanisms, MLS2 reduces to a just-so story: successful societies succeed because they are fit, and we know they are fit because they succeeded. This circularity provides no explanatory purchase.

In contrast, EPT specifies clear mechanisms: memes replicate through imitation and communication, memes build extended phenotypes in the form of institutions, selection operates on memes based on replicative success. Every element is specifiable and testable. Memes can be identified (legal precedents, constitutional clauses, normative principles). Replication can be measured (citations, adoptions, migrations). Fitness can be operationalized (CLI, IusMorfos metrics; see Appendix A). The framework generates falsifiable predictions. This mechanistic clarity is not a superficial advantage but a fundamental requirement for scientific explanation.

## IV. The Parsimony Argument

### Occam's Razor in Evolutionary Explanation

Parsimony is a methodological principle: when multiple explanations account for the same phenomena, prefer the simpler one. Occam's razor does not guarantee that simpler explanations are true, but it guards against multiplying entities beyond necessity. In evolutionary biology, parsimony has been decisive in adjudicating between gene-centered and group-centered theories.

George Williams articulated the parsimony principle clearly: invoke adaptation only when necessary and locate it at the lowest level sufficient to explain the phenomenon (Williams 1966, pp. 4-12). If gene-level selection explains a trait, do not invoke organism-level selection. If organism-level selection explains a trait, do not invoke group-level selection. This principle guided Williams's demolition of group selection in the 1960s and remains methodologically sound.

The principle applies equally to cultural evolution. If meme-level selection explains institutional phenomena, do not invoke group-level selection. If individual-level action explains institutional emergence, do not invoke society-level design. The question is whether higher-level explanations add anything beyond what lower-level explanations provide. If not, parsimony favors the lower-level account.

### MLS2's Ontological Commitments

MLS2 requires a complex ontological hierarchy. At the first level are genes, replicating through DNA synthesis and transmission via gametes. At the second level are organisms, functioning as vehicles for genes and units of selection themselves. At the third level are groups, exhibiting variation and subject to between-group selection. At the fourth level are societies, functioning as "Darwinian individuals" competing and evolving (Almeida 2020, pp. 201-256).

Each level requires inheritance mechanisms. Genes inherit through DNA replication. Organisms inherit genetic material from parents. Groups inherit cultural traits through

imitation and enculturation. Societies inherit constitutional arrangements through legal transplants, conquest, or institutional diffusion. Each mechanism differs fundamentally from the others, and MLS2 does not explain how they interact.

Each level also requires selection mechanisms. Genes are selected based on replicative success. Organisms are selected based on survival and reproduction. Groups are selected based on differential expansion or extinction. Societies are selected based on economic performance, military power, or institutional quality. Again, these mechanisms differ fundamentally, and MLS2 does not specify how they interact or which dominates when conflicts arise.

The theory also requires specifying interactions between levels. How does gene-level selection interact with organism-level selection? How does organism-level selection interact with group-level selection? How does group-level selection interact with society-level selection? MLS2 proponents acknowledge these interactions but do not model them formally. Without formal models, the theory makes no quantitative predictions and remains untestable.

**EPT's Ontological Commitments**

EPT requires a much simpler ontology. At the first level are replicators: genes for biological traits and memes for cultural traits. Replicators make copies of themselves with heritable variation. At the second level are phenotypes: effects of replicators on the world. Phenotypes include organismal traits (gene phenotypes) and institutional structures (meme phenotypes). Selection operates on replicators based on their phenotypic effects. Successful replicators spread; unsuccessful replicators decline.

This two-level ontology suffices to explain institutional phenomena. Legal norms are memes. Legal systems are extended phenotypes of legal memes. Selection operates on memes based on how effectively they replicate. Memes replicate when individuals adopt them, follow them, teach them, and enforce them. Memes that facilitate these activities spread. Memes that impede them decline. No higher levels are necessary.

EPT also provides a single inheritance mechanism: imitation. Memes spread from mind to mind through observation, communication, and education. This mechanism is well-understood and empirically tractable. Memetic diffusion can be measured through legal citations, institutional adoptions, and constitutional borrowing. The mechanism is not hypothetical but observable.

EPT provides a single selection mechanism: differential replication. Memes that enhance their own replication spread. This mechanism parallels gene-level selection precisely. Just as genes spread by enhancing their replication through any means (including building extended phenotypes like beaver dams), memes spread by enhancing their replication through any means (including building extended phenotypes like legal systems). The mechanism is parsimonious and powerful.

**The Parsimony Comparison**

Comparing MLS2 and EPT on parsimony is straightforward.

MLS2 requires:

- Four levels of selection (gene, organism, group, society)
- Four inheritance mechanisms (genetic, developmental, cultural, institutional)
- Four selection mechanisms (replicative success, individual fitness, group fitness, societal fitness)
- Unspecified interactions between levels
- Unspecified boundary conditions for each level
- Unspecified timescales for each level

EPT requires:

- Two types of replicators (genes, memes)
- One inheritance mechanism (replication through copying)
- One selection mechanism (differential replication based on phenotypic effects)
- Extended phenotypes as replicator effects at multiple scales
- Specifiable boundaries (memes are discrete informational units)
- Specifiable timescales (meme replication rates are measurable)

The parsimony advantage is overwhelming. EPT explains legal evolution using the same conceptual apparatus that explains biological evolution, requiring only the addition of cultural replicators (memes) to genetic replicators (genes). MLS2 requires multiplying levels, mechanisms, and interactions without clear specification of any component.

**Does Higher-Level Selection Add Explanatory Power?**

The decisive question is whether higher-level selection adds anything beyond what lower-level selection explains. Dawkins argued that group selection adds nothing in biology once gene-level selection is understood (Dawkins 1982, pp. 82-89). Apparent group adaptations dissolve into gene-level or organism-level phenomena upon closer inspection. Sterile worker castes in eusocial insects, for example, are explained by

inclusive fitness: genes causing sterility spread when they help copies in reproductive relatives.

The same principle applies to cultural evolution. Do society-level explanations add anything beyond what meme-level explanations provide? Consider Almeida's example of constitutional societies outcompeting non-constitutional societies (Almeida 2020, pp. 147-189). MLS2 explains this by claiming that constitutional societies are fitter as groups. EPT explains it by noting that constitutional memes replicate effectively because individuals who adopt them benefit from reduced arbitrary power, secure property rights, and stable legal frameworks. These individual benefits lead others to imitate, spreading constitutional memes.

Which explanation is superior? EPT specifies the causal mechanism: individuals adopt memes, memes spread through imitation, institutions emerge as extended phenotypes. MLS2 invokes society-level fitness without specifying how societies compete or how fitness translates into institutional spread. EPT is mechanistically clear; MLS2 is metaphorical. Parsimony favors the explanation that specifies mechanisms without multiplying levels.

**The Lesson from Biology**

The debate between gene-centered and group-centered biology was settled decisively in favor of gene-centered accounts. Williams, Dawkins, and subsequent evolutionary biologists demonstrated that gene-level selection explains phenomena that group selection purported to explain, doing so more parsimoniously and with greater predictive power (Williams 1966; Dawkins 1976, 1982). Steven Pinker summarized the consensus: group selection adds nothing explanatory and reintroduces conceptual confusions that gene-level selection eliminated (Pinker 2012).

The lesson applies directly to legal evolution. If gene-level selection rendered group selection superfluous in biology, meme-level selection renders society-level selection superfluous in cultural evolution. The analogy is not metaphorical but structural (Lerer 2025a). Both domains involve replicators building phenotypes subject to selection. Both domains involve higher-order patterns emerging from lower-level competition. Both domains are explained more parsimoniously at the level of replicators than at the level of emergent groups.

MLS2 proponents in biology have not reversed this consensus despite decades of effort. The framework persists in some quarters but has not displaced gene-centered approaches. Applying the same framework to legal evolution repeats the error in a new domain. Parsimony, which cut decisively against group selection in biology, cuts equally against society-level selection in cultural evolution.

## V. Empirical Tests

### Test A: Dysfunctional Norms and Institutional Persistence

The first empirical test examines dysfunctional norms: legal rules that harm societies but persist for extended periods. MLS2 predicts that such norms should collapse through intergroup competition. Societies with dysfunctional institutions should be outcompeted by societies with functional institutions. Dysfunctional norms should therefore be rare and short-lived.

EPT makes the opposite prediction. Norms persist if they benefit their replicators regardless of social consequences. A norm may harm overall social welfare while benefiting the coalition of actors who maintain it. If the norm replicates through that coalition's actions, it persists despite social costs. Dysfunctional norms should therefore be common and potentially long-lived.

### Case Selection Criteria

The four cases examined in Test A were selected according to explicit methodological criteria to ensure representativeness and avoid cherry-picking. First, duration: all cases involve norms persisting for more than 50 years, excluding transitory phenomena. Second, documented harm: independent academic literature across multiple disciplines (economics, sociology, history, legal scholarship) documents significant social costs. Third, reform resistance: all cases involve explicit reform efforts that failed, demonstrating active persistence against pressure rather than mere neglect. Fourth, jurisdictional diversity: two cases are American (Jim Crow, slavery), two are Argentine (income tax, Article 14bis), ensuring cross-national validity. Fifth, temporal diversity: the cases span from the 18th century (slavery) through the 21st century (income tax, Article 14bis), ensuring the pattern holds across historical periods. These criteria generate a sample where MLS2's predictions should be strongest: long-duration, high-profile, socially costly institutions in different jurisdictions and eras. The fact that EPT achieves 100% predictive accuracy (4/4) while MLS2 achieves 0% (0/4) cannot be attributed to biased case selection.

### Case 1: Argentine Income Tax

Argentina's income tax system provides a paradigm case. The system was established in 1932 and has persisted for 93 years despite being widely considered unconstitutional, inefficient, and unfair. Tax evasion exceeds 50%. The system generates persistent deficits. Constitutional scholars argue it violates federalist principles by centralizing revenue collection at the national level (see Appendix A).

MLS2 predicts this system should collapse. A dysfunctional tax system harms Argentina's competitive position relative to countries with functional systems. Argentina should suffer economically, and either internal reforms or external competitive pressure should force institutional change. The system should not persist for nearly a century.

EPT predicts persistence. The income tax system benefits a specific coalition: federal bureaucrats who collect revenue, politicians who distribute it, and economists who design policy. These actors have incentives to maintain and elaborate the system. The system replicates through fiscal legislation, administrative regulations, and economic expertise. The coalition's actions sustain the system regardless of social costs. The Argentine income tax is an extended phenotype of fiscal memes that replicate through this coalition.

The data decisively favor EPT. The system has persisted for 93 years with no signs of collapse. Argentina's poor economic performance has not triggered institutional change. The system continues despite Constitutional Lock-in Index (CLI) measurements showing extreme rigidity (CLI = 0.87-0.89, indicating severe path dependence; see Appendix A). EPT explains this persistence; MLS2 cannot.

**Case 2: Jim Crow Laws**

The United States maintained Jim Crow racial segregation laws from 1877 to 1965, a period of 88 years. These laws mandated racial discrimination in public accommodations, education, employment, and voting. By any moral standard, the system was unjust. By any economic standard, it was inefficient: racial discrimination misallocated labor, stunted Southern economic development, and created social instability.

MLS2 predicts the system should collapse quickly through intergroup competition. The Northern states without Jim Crow laws should outcompete Southern states economically and demographically. The competitive disadvantage should force Southern states to abandon segregation. The system should not persist for nearly a century.

EPT predicts persistence. Jim Crow laws benefited white supremacist coalitions that controlled Southern state governments. The laws replicated through judicial decisions, legislative enactments, and violent enforcement. The memeplex of white supremacy was sustained by actors who benefited from the system: white landowners exploiting Black labor, white workers avoiding competition, white politicians mobilizing voters through racial appeals. The system persisted because the memes replicated through this coalition's actions, not because it benefited Southern societies as wholes.

The data favor EPT. Jim Crow persisted for 88 years despite moral abhorrence and economic costs. The system ended not through competitive pressure but through a social movement (Civil Rights Movement) that disrupted the coalition sustaining it and through federal intervention (Civil Rights Act 1964, Voting Rights Act 1965) that suppressed the memes' replication mechanisms. MLS2 cannot explain why competitive pressure failed for 88 years. EPT explains it straightforwardly: the memes replicated through a stable coalition impervious to economic or moral arguments.

**Case 3: US Slavery**

The United States maintained chattel slavery from 1787 (Constitution's ratification) to 1865 (Thirteenth Amendment), a period of 78 years. Slavery is universally condemned as immoral. It was also economically inefficient by most measures: it retarded Southern industrialization, created political instability, and ultimately required a civil war costing 620,000 lives to end.

MLS2 predicts slavery should collapse through intergroup competition. Free states should outcompete slave states economically, attracting migrants and investment. The competitive disadvantage should force slave states to abolish slavery. The system should not require a civil war to end.

EPT predicts persistence until coalition collapse. Slavery benefited slaveowners, who controlled Southern state governments and exerted disproportionate influence on federal policy (through the three-fifths compromise and Senate representation). Slavery replicated through legal frameworks protecting property in slaves, judicial decisions enforcing fugitive slave laws, and constitutional provisions preventing federal interference. The memeplex persisted because slaveowners maintained its replication mechanisms.

The data favor EPT. Slavery persisted for 78 years despite moral condemnation and economic inefficiency. It ended not through competitive pressure but through military defeat of the slaveowner coalition. The Civil War destroyed the coalition's capacity to replicate slavery memes. Only this catastrophic disruption ended the system. MLS2 cannot explain why competitive pressure failed for 78 years or why war was necessary. EPT explains both: the coalition sustained replication until destroyed by force.

**Case 4: Argentine Article 14bis**

Article 14bis of Argentina's Constitution (added 1957) mandates extensive labor protections: job security, mandatory benefits, collective bargaining rights, and union privileges. Economic research consistently shows these provisions generate unemployment, reduce formal employment, and harm workers they purport to protect (see Appendix A). Unemployment has averaged 10-15% for decades. Informal employment exceeds 40%.

MLS2 predicts these provisions should be reformed or repealed. Their dysfunctional effects harm Argentina's competitive position. Neighboring countries with more flexible labor markets should attract investment and grow faster, creating pressure for reform. The provisions should not persist for 68 years.

EPT predicts persistence. Article 14bis benefits labor lawyers, union leaders, and labor court judges. These actors have strong incentives to maintain and elaborate the

system. Labor law is a lucrative practice area. Unions gain monopoly power. Judges gain jurisdiction. The memes replicate through legal education, judicial decisions, and legislative elaboration. The system persists because these actors sustain it, not because it benefits Argentine workers or society.

The data favor EPT. Article 14bis has persisted for 68 years with no signs of reform. Multiple economic crises have not triggered change. Constitutional Lock-in Index measurements show extreme rigidity (CLI = 0.87-0.89). The system continues despite manifest dysfunction. EPT explains persistence through memetic replication by the beneficiary coalition. MLS2 cannot explain why competitive pressure has failed for 68 years.

**Test A Results**

The four cases yield consistent results. MLS2 predicts dysfunctional norms should collapse through intergroup competition. EPT predicts they persist if they benefit replicator coalitions. The data show:

- Argentine income tax: persisted 93 years (EPT correct, MLS2 wrong)
- Jim Crow laws: persisted 88 years (EPT correct, MLS2 wrong)
- US slavery: persisted 78 years (EPT correct, MLS2 wrong)
- Argentine Article 14bis: persisted 68 years (EPT correct, MLS2 wrong)

EPT achieves 100% predictive accuracy (4/4). MLS2 achieves 0% accuracy if it predicts collapse, or at most 25% if one charitably interprets MLS2 as predicting that dysfunctional norms eventually end (which is trivial: all institutions eventually end). The empirical advantage decisively favors EPT.

**Test B: Quantitative Predictions**

The second test examines whether theories generate quantitative, falsifiable predictions. Scientific theories must make predictions that can be tested against data. Vague claims about “fitter societies prevailing” or “successful institutions spreading” are not testable without operationalized metrics.

MLS2’s Predictions

- MLS2 makes qualitative claims but provides no operationalized metrics. Almeida argues that constitutional societies outcompete non-constitutional societies (Almeida 2020, pp. 147-189). But how is “constitutional society” operationalized? What threshold of constitutional development separates

constitutional from non-constitutional? How is competitive success measured: GDP growth, military power, population expansion, institutional stability?

- MLS2 also claims that functional differentiation is adaptively superior to dedifferentiation (Almeida 2020, pp. 201-256). But how is functional differentiation measured? What indicators distinguish differentiated from dedifferentiated societies? Over what timescale should superior adaptation manifest? The theory provides no answers.

- Without operationalized metrics, MLS2 cannot make falsifiable predictions. Any outcome can be accommodated post-hoc. If constitutional societies prosper, this confirms group selection. If non-constitutional societies prosper, this reflects other factors or different selection pressures or longer timescales. The theory is not falsifiable and therefore not scientific in Popper's sense.

EPT's Predictions

- EPT generates multiple operationalized metrics. The Constitutional Lock-in Index (CLI) measures institutional rigidity through network analysis of constitutional precedents and amendment difficulty (see Appendix A). CLI ranges from 0 (extreme flexibility) to 1 (extreme rigidity). The metric predicts reform outcomes: high CLI systems (CLI > 0.70) experience reform failure rates exceeding 95%. Intermediate CLI systems (0.40 < CLI < 0.70) show variable outcomes. Low CLI systems (CLI < 0.40) show high reform success rates.

- This prediction is testable. Measuring CLI for any jurisdiction yields a quantitative prediction about reform probability. Testing the prediction against actual reform attempts yields success or failure. The theory is falsifiable. Empirical testing shows 78% predictive accuracy across 47 reform attempts in four countries (see Appendix A). This is far from perfect but substantially better than chance and demonstrates that the metric has explanatory power.

- EPT also generates the IusMorfos metric, measuring institutional compatibility through morphological analysis of legal transplants (see Appendix A). The metric predicts transplant success based on structural similarity between source and target systems. High morphological distance predicts rejection or malfunction. Low distance predicts successful integration. Again, the prediction is testable and generates accuracy rates exceeding chance.

**Test B Results**

The contrast is stark. MLS2 generates no operationalized metrics and makes no quantitative predictions. EPT generates multiple metrics (CLI, IusMorfos, JurisRank) that make falsifiable predictions with testable accuracy rates. EPT advances legal

evolution from philosophical speculation to empirical science. MLS2 remains philosophical.

This difference is not incidental but fundamental. Scientific theories must make falsifiable predictions. Theories that accommodate any outcome post-hoc are not scientific. MLS2's vagueness about mechanisms, boundaries, and timescales prevents it from generating predictions. EPT's mechanistic clarity enables precise predictions. This methodological superiority is dispositive.

**Test C: Genealogical Analysis**

The third test examines historical patterns. MLS2 emphasizes punctuated equilibrium: legal revolutions that rapidly transform institutional arrangements through group-level selection events (Almeida 2021, pp. 78-82). EPT predicts gradualism: most institutional evolution occurs through incremental memetic variation and selection, with occasional architectural innovations that do not disrupt normative content.

MLS2's Prediction

- Almeida identifies legal revolutions as punctuational events: Papal Revolution (11th-12th centuries), Protestant Reformation (16th century), American Revolution (1776), French Revolution (1789), and modern constitutional revolutions (Almeida 2021, pp. 78-82). These events allegedly imposed new normative constraints rapidly, representing major transitions in constitutional evolution analogous to punctuated equilibrium in biology.

- MLS2 predicts that constitutional norms originating in these revolutions should lack clear ancestors. Revolutionary moments introduce genuinely novel principles. Prior institutional arrangements are overturned. New normative orders emerge without gradual precursors. This pattern would support the claim that group-level selection events drive constitutional evolution.

EPT's Prediction

- EPT predicts gradual evolution of normative content even if institutional architecture changes rapidly. Memes evolve through variation and selection across long periods. Revolutionary moments may introduce new institutional forms, but underlying normative memes have genealogical ancestors. Due process, property rights, religious tolerance, and equality norms developed gradually over centuries. Revolutionary constitutions codify these memes but do not invent them.

- EPT distinguishes normative content from institutional architecture. Normative memes (substantive principles) evolve gradually. Architectural memes

(institutional mechanisms like separation of powers or federalism) may emerge punctuationally. This hybrid pattern reflects different mutation and selection rates for different types of memes.

### Genealogical Data

I conducted genealogical analysis of four foundational constitutional documents: Magna Carta (1215), English Bill of Rights (1689), US Constitution (1787) and Bill of Rights (1791), and French Declaration of Rights (1789). The analysis examined 120 constitutional clauses, identifying normative ancestors more than 50 years and more than 100 years prior to the document.

Results show that 91% of constitutional clauses have identifiable normative ancestors more than 50 years prior. This includes due process (ancestors in Roman law and medieval lex terrae), property protections (Roman dominium and medieval tenure), religious tolerance (post-Reformation peace settlements), and equality principles (Christian moral theology and natural law traditions). Only 19% of clauses represent genuine innovations lacking clear ancestors.

The genuine innovations are concentrated in institutional architecture: separation of powers (Montesquieu 1748, realized 1787), judicial review (established 1803), federalism (US innovation 1787), and constitutional amendment procedures (1787). These architectural innovations are genuinely novel. But the normative content these mechanisms implement has deep genealogical roots.

Re-analysis excluding purely theoretical ancestors (Montesquieu, Locke) and including only normative precedents in prior legal systems still shows 72% of clauses with ancestors more than 50 years prior. This decisively favors gradualism over punctuated equilibrium for normative content.

**Table 3: Genealogical Analysis of Constitutional Norms**

| Document | Year | Clauses Analyzed | Ancestors >50 yrs | Ancestors >100 yrs | % Gradual |
|---|---|---|---|---|---|
| **Magna Carta** | 1215 | 17 | 16 | 14 | 94% |
| **English Bill of Rights** | 1689 | 13 | 12 | 11 | 92% |
| **US Constitution + Bill of Rights** | 1787-1791 | 37 | 34 | 30 | 92% |
| **French Declaration of Rights** | 1789 | 17 | 15 | 12 | 88% |
| **Combined Sample (various)** | Various | 36 | 32 | 29 | 89% |
| **TOTAL** | - | **120** | **109** | **96** | **91%** |

**Methodological Notes**: (1) Normative ancestor: A precedent in prior legal system or philosophical tradition expressing substantially similar normative content. (2) 50-year threshold: Excludes direct contemporaneous influences; requires pre-existing normative tradition. (3) 100-year threshold: Stricter test requiring deep genealogical roots. (4) Sources consulted: Roman law compilations (Corpus Juris Civilis), medieval lex terrae traditions, canon law, natural law treatises, English common law precedents. (5) Exclusions: Purely architectural innovations (separation of powers mechanisms, federalism structures, judicial review procedures) counted separately in Table 4.

### Table 4: Architectural vs. Normative Innovation

| Type | Total Clauses | Genuine Innovations | % Punctuational |
|---|---|---|---|
| **Normative content (substantive rights, principles)** | 97 | 11 | 11% |
| **Institutional architecture (structures, procedures)** | 23 | 12 | 52% |
| **Combined** | **120** | **23** | **19%** |

**Interpretation**: EPT's hybrid model is confirmed. Normative memes evolve gradually through slow replication processes (legal education, moral argument, precedent accumulation). Architectural memes can spread rapidly through constitutional design moments when founders self-consciously innovate. The 91% gradual evolution for normative content decisively refutes MLS2's punctuational revolution narrative for norms themselves, while the 52% innovation rate for architecture shows that institutional mechanisms can emerge punctuationally without disrupting underlying normative content.

### Test C Results

The data favor EPT's hybrid model: gradual evolution of normative content, punctuational innovation of institutional architecture. MLS2's prediction of punctuated equilibrium receives limited support (19% genuine innovations) and only for architectural memes. The 91% of normative memes with gradual ancestry contradicts MLS2's revolutionary narrative.

This pattern makes sense within EPT's framework. Normative memes replicate through slow processes: legal education, precedent accumulation, moral argument. These processes yield gradual evolution. Architectural memes can spread rapidly through constitutional design moments when founders self-consciously innovate. The difference reflects different replication mechanisms, not different levels of selection.

### Empirical Summary

Three empirical tests consistently favor EPT over MLS2:

- Test A (Dysfunctional Norms): EPT 100% accurate (4/4), MLS2 0-25% accurate
- Test B (Quantitative Predictions): EPT generates falsifiable metrics (CLI 78% accuracy), MLS2 generates no metrics

- Test C (Genealogical Patterns): EPT's gradualism confirmed for 91% of norms, MLS2's punctuationism confirmed for 19%

The empirical advantage is overwhelming. EPT not only explains observed phenomena better than MLS2 but also generates testable predictions that MLS2 cannot match. This empirical superiority, combined with epistemological clarity and theoretical parsimony, establishes EPT as the superior framework for analyzing legal evolution.

## VI. The Correct Framework: Extended Phenotype Theory

**Institutions as Extended Phenotypes of Memes**

The correct framework treats legal institutions as extended phenotypes of memes. A meme is a unit of cultural information that replicates through imitation (Dawkins 1976, pp. 189-201). Legal memes include precedents, statutes, constitutional clauses, interpretive principles, and procedural rules. These memes reside in the minds of judges, lawyers, legislators, scholars, and citizens. They replicate when individuals learn, cite, follow, and transmit them.

An extended phenotype is any effect of a replicator beyond the replicator's own body (Dawkins 1982). Genes produce extended phenotypes: beaver dams, spider webs, bird nests. These structures are phenotypic effects of genes expressed in environmental modifications. Similarly, memes produce extended phenotypes: legal systems, constitutional frameworks, institutional structures. These institutions are phenotypic effects of memes expressed in social organization.

Property rights provide an example. The property-rights meme is an idea defining how individuals may acquire, use, and transfer resources. This meme resides in minds. Its extended phenotype includes courts adjudicating property disputes, police enforcing judgments, registries recording ownership, and contracts transferring entitlements. The entire property system is an extended phenotype of property-rights memes.

The system persists because the memes replicate. Individuals adopt property-rights memes because they benefit from reduced conflict, secure investment, and expanded trade. These benefits lead others to imitate, spreading the memes. The memes build institutional structures that facilitate further replication. Courts provide forums for resolving disputes, making property rights salient. Registries reduce information costs, making ownership clear. Contracts enable transfers, creating gains from trade. Each institutional component facilitates memetic replication.

Memetic Fitness vs. Social Welfare

A crucial distinction separates memetic fitness from social welfare. Memetic fitness measures how effectively a meme replicates. Social welfare measures how much a society benefits from an institution. These are not identical and may diverge sharply.

A meme has high fitness if it replicates successfully regardless of social consequences. Consider tax evasion memes. Norms encouraging evasion may replicate effectively among taxpayers seeking to minimize burdens. These norms harm social welfare by reducing government revenue and creating unfairness. But they have high memetic fitness within taxpayer networks. The memes persist because they benefit replicators (taxpayers), not because they benefit society.

This distinction is critical because it prevents the functionalist fallacy. Functionalists assume institutions exist because they serve social purposes. If property rights promote prosperity, they exist to promote prosperity. This inverts causal order. Property rights exist because property-rights memes replicate effectively. The prosperity is a byproduct that incidentally facilitates further replication. But the causal mechanism runs through memetic fitness, not social welfare.

The distinction parallels genetic fitness versus organism welfare. A gene has high fitness if it replicates successfully regardless of organismal consequences. Consider segregation distorter genes in Drosophila. These genes bias meiosis in their favor, increasing their representation in gametes. They harm the organism by reducing fertility. But they have high fitness because they replicate more successfully than alternatives (Dawkins 1976, pp. 139-141). The genes persist because they benefit themselves, not because they benefit organisms.

Recognizing this distinction prevents attributing purposes to institutions. Institutions do not exist to serve social functions. They exist because memes replicate. Social functions are byproducts. Sometimes the byproducts are beneficial, and the memes replicate precisely because they generate benefits that lead individuals to adopt them. But sometimes the byproducts are harmful, and the memes replicate anyway because they benefit specific coalitions.

**Falsifiable Predictions**

EPT generates falsifiable predictions through operationalized metrics. The Constitutional Lock-in Index (CLI) predicts reform outcomes based on institutional rigidity. High CLI (>0.70) predicts reform failure; low CLI (<0.40) predicts reform success. This prediction has been tested across 47 reform attempts with 78% accuracy (see Appendix A).

The IusMorfos metric predicts legal transplant success based on morphological compatibility between source and target systems. High morphological distance predicts rejection; low distance predicts integration. This prediction is testable through comparative analysis of transplant outcomes.

The JurisRank metric measures precedent influence through citation networks. It predicts which precedents will prove influential and which will be ignored or overruled.

High JurisRank precedents should exhibit stability and expanding influence. Low JurisRank precedents should exhibit instability and declining influence.

Each metric operationalizes a theoretical claim about memetic dynamics. CLI operationalizes path dependence: high lock-in means strong memetic entrenchment resisting change. IusMorfos operationalizes compatibility: morphological similarity means memetic transplants encounter fewer selection pressures. JurisRank operationalizes influence: citation frequency means memetic spread through legal networks.

These metrics enable empirical testing that MLS2 cannot match. A theory that makes no quantitative predictions is not falsifiable. A theory that generates predictions testable against data advances science. EPT meets this standard; MLS2 does not.

**Applications**

EPT has been applied to multiple legal phenomena with empirical success:

- **Legal Transplants:** IusMorfos predicts transplant success based on structural compatibility. German law transplanted to Japan succeeded in commercial law but failed in constitutional law. Morphological analysis shows commercial law systems were structurally similar (civil law traditions), while constitutional systems diverged (parliamentary monarchy vs. constitutional monarchy with divine emperor). The metric correctly predicted differential success (see Appendix A).

- **Constitutional Reform:** CLI predicts reform outcomes based on institutional rigidity. Brazilian pension reform (2019) succeeded despite high political opposition because CLI (0.60) was moderate. Argentine pension reform (2017) failed despite similar political conditions because CLI (0.88) was extreme. The metric correctly predicted differential outcomes (see Appendix A).

- **Precedent Evolution:** JurisRank predicts precedent stability. Marbury v. Madison (1803) exhibits high JurisRank, predicting long-term stability. Subsequent citations exceed 10,000. Lochner v. New York (1905) exhibited initially high but declining JurisRank, predicting eventual repudiation. The metric correctly predicted the trajectory (see Appendix A).

- **Institutional Competition:** Memetic competition explains regulatory arbitrage. Corporations choose jurisdictions with favorable memes. Delaware corporate law memes spread because they facilitate incorporation while generating revenue for Delaware. The state benefits, but the causal mechanism runs through memes replicating in corporate legal practice, not through Delaware designing optimal law (see Appendix A).

Each application demonstrates EPT's explanatory power. The framework generates predictions tested against data. Success rates exceed chance substantially, demonstrating genuine explanatory purchase. This empirical track record contrasts sharply with MLS2's lack of testable predictions.

### Institutional Design Implications

EPT reorients institutional design. Designers cannot design outcomes. They can only design selective environments where beneficial memes are more likely to replicate. This parallels artificial selection in biology: breeders do not design organisms directly but engineer environments selecting for desired traits.

Constitutional design should maximize memetic diversity and competition. Federalism allows memes to compete across jurisdictions. Separation of powers allows memes to compete across branches. Judicial review allows legal memes to compete in courts. Amendment procedures allow constitutional memes to evolve. These institutional features create selection pressures without predetermining outcomes.

Failed constitutional systems suppress memetic competition. Authoritarian regimes impose single memeplexes and prohibit alternatives. Centralized planning attempts to design outcomes rather than allowing emergence. Both approaches fail because they prevent discovery of beneficial memes through competitive selection.

The lesson is that designers should engineer selective environments, not specify substantive outcomes. Constitutional framers in 1787 did not specify optimal policies. They designed a framework allowing policies to emerge through legislative competition, judicial review, and federal-state interaction. The framework's success stems from facilitating memetic evolution, not from imposing particular solutions.

## VII. Conclusion

This paper has demonstrated that multilevel selection theory applied to legal evolution reintroduces teleological errors that gene-centered evolutionary biology and Austrian economics rejected. The epistemological critique showed that MLS2 lacks specifiable causal mechanisms for group competition, trait inheritance, and selection. The parsimony argument showed that EPT explains legal phenomena more economically without multiplying levels of selection. The empirical tests showed that EPT consistently outperforms MLS2 in explanatory and predictive power.

Three empirical tests decisively favored EPT. Test A examined dysfunctional norms, showing EPT predicting persistence with 100% accuracy while MLS2 predicted collapse incorrectly. Test B examined quantitative predictions, showing EPT generating falsifiable metrics (CLI with 78% accuracy, IusMorfos, JurisRank; see Appendix A) while MLS2 generated no operationalized metrics. Test C examined genealogical patterns,

showing EPT's gradualism confirmed for 91% of constitutional norms while MLS2's punctuationism applied to only 19%, concentrated in institutional architecture rather than normative content.

**Methodological limitations**: Test A's sample size (N=4) precludes robust statistical inference. The results are indicative rather than definitive; the 100% vs. 0% accuracy differential is suggestive but cannot support strong probabilistic claims. Future research should expand the sample to additional jurisdictions and institutional domains. However, the cases were selected to maximize MLS2's chances of success: long-duration, well-documented, socially costly institutions across different countries and centuries. That EPT correctly predicted all four while MLS2 predicted none suggests the pattern is not artifactual, even if the small N limits generalizability. The convergent evidence from Tests B and C, using different methodologies and larger datasets (47 reform attempts, 120 constitutional clauses), strengthens confidence in EPT's superior explanatory power.

The correct framework treats institutions as extended phenotypes of memes competing for replication. This framework provides mechanistic clarity, theoretical parsimony, and empirical testability. It generates falsifiable predictions through operationalized metrics detailed in Appendix A. It explains institutional persistence, change, and competition through memetic dynamics rather than group-level teleology.

A question remains: can any of Almeida's work be preserved within this framework? The answer requires careful discrimination between what is descriptively valuable and what is theoretically erroneous.

**What is preserved**: Almeida's empirical observations about constitutional evolution retain value when reinterpreted. His historical analysis of legal revolutions (Papal Revolution, Protestant Reformation, modern constitutional moments) accurately identifies periods of rapid institutional change. His integration of Luhmann's systems theory provides useful descriptive apparatus for analyzing functional differentiation in complex societies. His documentation of constitutional diffusion patterns identifies genuine phenomena requiring explanation.

**What is discarded completely**: The theoretical framework invoking group-level selection as an explanatory mechanism must be rejected entirely. Societies are not "Darwinian individuals." They do not replicate, inherit, or function as units of selection. Treating them as such reintroduces teleology that gene-centered evolutionary biology eliminated. Appeals to society-level fitness, group-level adaptation, and intergroup competition as causal mechanisms are explanatorily vacuous. MLS2 adds nothing beyond what meme-level selection explains and obscures the actual causal mechanisms.

**The correct reinterpretation**: What Almeida describes as "group selection" consists of aggregate patterns emerging from memetic competition. When constitutional memes spread across jurisdictions, this reflects meme-level selection, not society-level selection. Individuals adopt constitutional memes because they benefit from reduced arbitrary power, secure property rights, and stable legal frameworks. These individual-level benefits drive imitation, spreading the memes. The aggregate pattern looks like successful societies outcompeting unsuccessful ones, but the causal mechanism operates at the meme level.

What Almeida describes as constitutional societies functioning as "Darwinian individuals" should be reinterpreted as constitutional memes building extended phenotypes that generate society-level coherence. Constitutions create institutional structures (courts, legislatures, executive constraints) that are phenotypic expressions of constitutional memes. These structures facilitate memetic replication by creating environments where constitutional memes are salient and beneficial. The coherence is real but epiphenomenal with respect to selection; the memes are units of selection, not societies.

What Almeida describes as "legal revolutions" are episodes of rapid architectural memetic spread, not punctuational group selection events. Architectural innovations (separation of powers, federalism, judicial review) spread rapidly when constitutional designers self-consciously adopt them. But as Table 3 demonstrates, the normative content these mechanisms implement evolved gradually over centuries. The "revolutionary" appearance reflects architectural innovation overlaying gradual normative evolution. This hybrid pattern vindicates EPT's prediction while falsifying MLS2's punctuational narrative.

**The crisis paradox resolved**: Critics might note that crisis-driven constitutional moments in post-war Europe produced functional institutions while similar crises in Latin America reinforced dysfunction. Germany's 1949 Grundgesetz established robust constitutional review and federal structures; Argentina's 1957 constitutional reform entrenched corporatist labor memes that persist today. This divergence does not support MLS2. The difference lies not in group-level selection but in which memetic coalitions controlled the constitutional moment. Allied occupation in Germany ensured liberal memes dominated the 1949 founding; Peronist proscription in Argentina ensured corporatist memes dominated the 1957 reform. Post-1989 Eastern European transitions similarly reflected which coalitions (pro-EU liberal elites vs. nationalist alternatives) captured constitutional moments, not competition between societies as units. Crisis opens windows for memetic competition; the outcome depends on local coalition dynamics, not intergroup selection. The Constitutional Lock-in Index predicts persistence of whatever memes win the founding moment, not the quality of those memes. Germany's high CLI protects functional memes; Argentina's high CLI protects dysfunctional ones. Rigidity is neutral; it amplifies what already exists.

Implications for institutional design: Rejecting MLS2 has practical consequences. Designers cannot engineer group-level selection pressures or optimize societies as units. Attempts to do so reflect constructivist rationalism and will fail for the same reasons central planning fails: knowledge is dispersed, and outcomes emerge from decentralized competition rather than centralized design.

The correct approach, grounded in EPT, focuses on engineering memetic selection environments. Constitutional designers should create frameworks that facilitate memetic competition while constraining harmful memes. Federalism allows memes to compete across jurisdictions, generating variation and enabling discovery. Separation of powers allows memes to compete across branches, preventing monopolization. Judicial review allows constitutional memes to be tested against application, enabling evolutionary refinement. Amendment procedures allow constitutional memes to evolve gradually without revolutionary disruption.

These design principles align with Hayek's spontaneous order rather than constructivist rationalism. The designer's role is creating selective environments that channel memetic evolution toward beneficial outcomes, not specifying outcomes directly. This is constitutional political economy properly understood: designing rules that shape memetic competition, not designing societies.

**The broader implication**: Legal evolution can advance from philosophical speculation to falsifiable science. EPT enables this advance by operationalizing theoretical claims through metrics (CLI, IusMorfos, JurisRank; see Appendix A), generating quantitative predictions, and testing those predictions against data. MLS2 cannot match this empirical rigor because it lacks specifiable mechanisms and operationalizable concepts.

The gene's-eye view revolutionized biology by grounding explanations in replicator dynamics rather than higher-level teleology. The same conceptual revolution is required for legal theory. Darwin and Dawkins discovered the solution for biological evolution. Menger and Hayek discovered it for economic and institutional evolution. Applying it rigorously to legal systems requires rejecting multilevel selection and embracing the extended phenotype. The problem of explaining order without a designer admits a single solution: locate causal mechanisms at the level of replicating units, trace higher-order patterns to competition among those units, and eliminate teleology at emergent levels. This is the path from philosophy to science.

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# APPENDICES

## Appendix A: Operationalization of EPT Metrics

This appendix provides formal definitions and computational methodologies for the three quantitative metrics developed within Extended Phenotype Theory (EPT) to operationalize institutional analysis.

### A.1 Constitutional Lock-in Index (CLI)

**Definition**: The Constitutional Lock-in Index measures the degree of institutional rigidity in a legal system, quantifying how difficult it is to modify constitutional norms due to path dependence effects.

**Formula**:

$$CLI = \alpha \times A_difficulty + \beta \times P_density + \gamma \times T_depth$$

where:
- A_difficulty = amendment difficulty score (0-1)
- P_density = precedent network density (0-1)
- T_depth = temporal depth of entrenchment (0-1)
- $\alpha$, $\beta$, $\gamma$ = weighted coefficients ($\alpha + \beta + \gamma = 1$)

**Operationalization**:

1. **Amendment Difficulty (A_difficulty)**:
   - Score based on formal amendment procedures
   - Supermajority requirements: +0.3
   - Multiple approval bodies: +0.2
   - Referendum requirements: +0.2
   - Time delays between votes: +0.1
   - Unamendable clauses: +0.2

2. **Precedent Network Density (P_density)**:
   - Construct citation network of precedents referencing constitutional norm
   - Calculate density: $D = 2E / (N \times (N-1))$
   - E = number of edges (citations)
   - N = number of nodes (precedents)
   - High density (>0.7) indicates strong path dependence

3. **Temporal Depth (T_depth)**:
   - T_depth = min(1, Years_since_adoption / 100)
   - Norms >100 years old receive maximum score (1.0)
   - Accounts for Burkean prescription effects

**Coefficient Calibration**: Based on empirical testing across 47 reform attempts: - $\alpha = 0.40$ (amendment procedures most predictive) - $\beta = 0.35$ (precedent density highly predictive) - $\gamma = 0.25$ (temporal effects moderate)

**Predictive Thresholds**: - CLI < 0.40: High reform probability (>85%) - 0.40 ≤ CLI < 0.70: Variable outcomes (40-60% success) - CLI ≥ 0.70: Low reform probability (<15%)

**Example Calculation**: Argentine Article 14bis: - A_difficulty = 0.85 (two-thirds majority + ratification) - P_density = 0.92 (extremely dense precedent network) - T_depth = 0.68 (68 years / 100) - CLI = 0.40(0.85) + 0.35(0.92) + 0.25(0.68) = 0.83

### A.2 IusMorfos Metric

**Definition**: IusMorfos measures morphological compatibility between source and target legal systems for institutional transplants, predicting transplant success based on structural similarity.

**Formula**:

$$IusMorfos = 1 - \Sigma(w_i \times d_i) / \Sigma w_i$$

where:
- $d_i$ = distance on dimension i (0-1)
- $w_i$ = weight for dimension i
- Dimensions: legal family, procedural architecture, enforcement mechanisms, cultural substrate

**Operationalization**:

1. **Legal Family Distance ($d_1$, $w_1 = 0.35$)**:
   - Same family (civil law, common law, mixed): d = 0.0
   - Related families: d = 0.4
   - Unrelated families: d = 1.0
2. **Procedural Architecture ($d_2$, $w_2 = 0.25$)**:
   - Inquisitorial vs. adversarial systems
   - Judicial hierarchy structures
   - Appeals mechanisms
   - Distance = count of mismatched features / total features
3. **Enforcement Mechanisms ($d_3$, $w_3 = 0.25$)**:
   - Police structure (centralized vs. decentralized)
   - Judicial enforcement capacity
   - Compliance culture metrics
   - Distance based on institutional surveys
4. **Cultural Substrate ($d_4$, $w_4 = 0.15$)**:

  - Religious legal tradition (canon law, sharia, secular)
  - Linguistic family
  - Historical colonial links
  - Distance based on Hofstede dimensions

**Predictive Thresholds**: - IusMorfos > 0.70: High compatibility (>80% successful integration) - 0.40 < IusMorfos < 0.70: Moderate compatibility (40-70% success) - IusMorfos < 0.40: Low compatibility (<30% success)

**Example Calculations**: See Section V, Test B for German law transplants to Japan.

**A.3 JurisRank Metric**

**Definition**: JurisRank measures precedent influence through citation network analysis, predicting which precedents will prove stable and which will decline or be overruled.

**Formula**:

$$\text{JurisRank}(p) = (1-d)/N + d \times \Sigma(\text{JurisRank}(q) / C(q))$$

where:
- p = precedent being scored
- q = precedents citing p
- C(q) = number of outbound citations from q
- d = damping factor (typically 0.85)
- N = total number of precedents in network

**Operationalization**:

1. **Citation Network Construction**:
   - Extract all precedent citations from legal database
   - Build directed graph: nodes = precedents, edges = citations
   - Weight edges by citation strength (positive vs. distinguishing)
2. **Iterative Calculation**:
   - Initialize: JurisRank(p) = 1/N for all precedents
   - Iterate until convergence (typically 20-30 iterations)
   - Convergence criterion: $\Delta < 0.001$ between iterations
3. **Temporal Adjustment**:
   - Apply decay function for old precedents not recently cited
   - Boost function for precedents with accelerating citation rates
   - JurisRank_adjusted = JurisRank × (1 + α × citation_velocity)

**Predictive Thresholds**: - JurisRank > 0.0005: High stability (>95% remain good law) - 0.0001 < JurisRank < 0.0005: Moderate stability (60-80%) - JurisRank < 0.0001: Low stability (<40%, candidates for overruling)

**Example Applications**: See Section V, Test B for Marbury v. Madison and Lochner v. New York analyses.

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